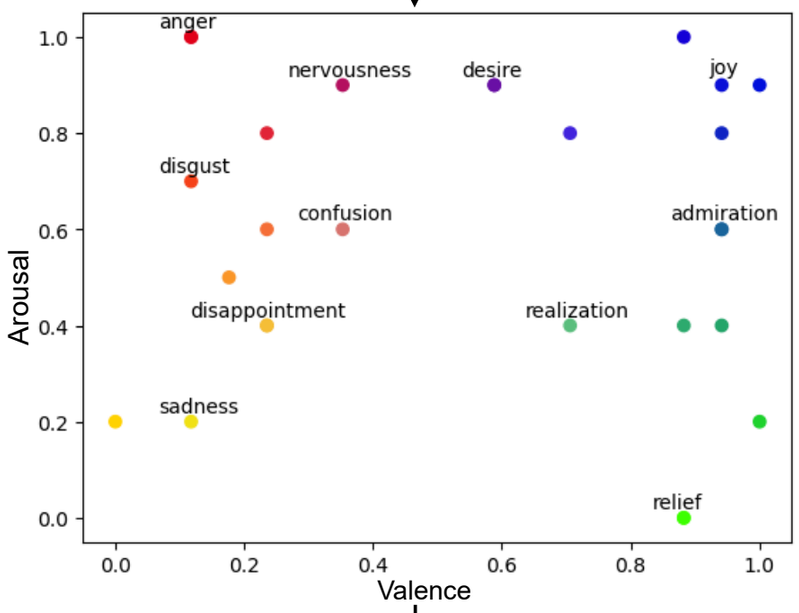
## I. Global Step: Cluster label space and create BWS tuples

**Emotion labels available**

admiration | amusement | anger | annoyance | approval | ... | pride | relief | remorse | sadness | surprise

*Query LLM for Valence/Arousal values*

LLM




*Cluster/Partition emotions into BWS tuples*

**BWS tuples of size *M***

**Valence, similar**

admiration, amusement, gratitude, joy, neutral | anger, disgust, grief, fear, neutral

**Valence, heterogeneous**

admiration, fear, realization, nervousness, neutral | anger, approval, confusion, joy, neutral

**Arousal, similar**

admiration, annoyance, confusion, optimism, neutral | anger, excitement, fear, surprise, neutral

**Arousal, heterogeneous**

admiration, anger, amusement, nervousness, neutral | annoyance, curiosity, excitement, grief, neutral

*Fill Prompt Template*

## II. Inference: LLM BWS scoring + random walks on *worse-than* graph

**Input text**

*Awesome! It was really interesting to hear your experience. Thanks for sharing.*

*Fill Prompt Template*

**Prompt template**

You are an expert annotator specializing in emotion recognition.

Please consider the Reddit post and the candidate emotions listed below.
Emotions: ***{bws tuple}***
Text: ***{text}***
From the given emotions, select both the most likely and the least likely emotion that the author of the Reddit post may have felt.

Only return the names of the most and least likely emotion.

*Prompt LLM with every BWS tuple*

LLM

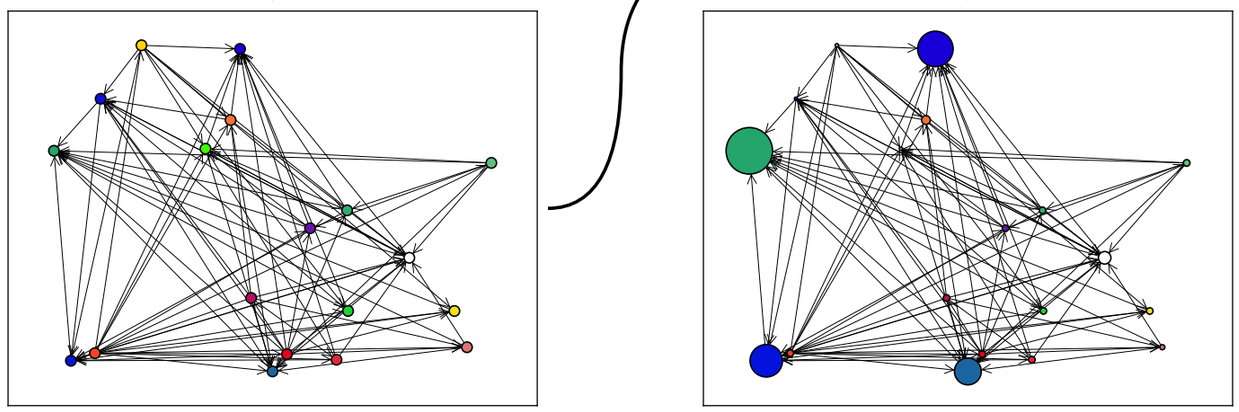
**BWS tuples evaluated by LLM**

WORST: amusement < BEST: {admiration, joy, neutral} < gratitude
...
WORST: anger < BEST: {approval, confusion, neutral} < joy

WORST: annoyance < BEST: {confusion, optimism, neutral} < admiration
...
WORST: grief < BEST: {annoyance, curiosity, neutral} < excitement

*Build graph from all "<"-relations*

*Random Walks*



**Final scores per emotion**

